Institute of Information Technology
University of Dhaka

Topic: Goal Question Metrics (GQM)

Software Metrics (SE 611)

Submitted to:

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1. Introduction

GQM is the abbreviation for "goal, question, metric". It is an established goal-oriented approach to software metrics to improve and measure software quality.

GQM defines a measurement on three dimensions. They are-

Conceptual Level (Goal): A goal is defined for an object, for a variety of reasons,with respect to various models of quality, from various points of view and relative to a particular environment.

Operational level (Question): A set of questions is used to define models of the object of study and then focuses on that object to characterize the assessment or Achievement of a specific goal.

Quantitative level (Metric): A set of metrics, based on the models, is associated With every question in order to answer it in a measurable way.

The first step is deciding our goal, then develop a question set to collect data from stakeholders. The questions must have appropriate metrics to
measure the result later. We have to create a questionnaire to collect data from
people.
defined in the scope. The last part will include metrics analysis with result derivation to finally reach an outcome.

2. Project Specifications

2.1 Overview

Mental health is a critical factor in a student's academic performance and overall well-being. Software measurement, especially with the Goal-Question-Metric (GQM) approach, helps systematically assess such intangible aspects.

2.2 Motivation

The GQM approach allows for a structured investigation into mental health issues by defining clear goals, deriving questions, and establishing measurable metrics. This ensures that the subjective and sensitive nature of mental health is explored rigorously.

2.3 Scope

This assignment focuses on identifying and evaluating the societal and personal factors that impact the mental health of IIT DU students. Through survey responses, key trends and patterns will be analyzed to support institutional improvements.

3. Goal Specification

3.1 GQM Framework

The GQM approach consists of three levels: Goal (what you want to achieve), Question (what you need to ask to know if you're achieving the goal), and Metric (what data is used to answer the questions). It is particularly useful in contexts like this where qualitative and quantitative factors interact.

General Statement: Investigating the societal and personal factors contributing to the deterioration of mental health among IIT DU students by measuring the frequency of academic stress, lifestyle patterns, institutional support, and student-identified stressors.

3.2 PPE Approach

Purpose: To evaluate how personal responsibilities, academic challenges, and institutional limitations contribute to the mental health condition of IIT DU students.

Perspective: To understand the lived experiences of students from different academic years, gender identities, and socio-economic backgrounds in relation to mental well-being.

Environment: This study focuses on students currently enrolled in the Institute of Information Technology (IIT), University of Dhaka, who reside in diverse environments including on-campus, off-campus, and at home, and who are navigating varying academic and social pressures.

After specifying our goal through the purpose-perspective-environment framework, our final goal is -

"To explore and evaluate the personal and societal stressors that contribute to poor mental health among IIT DU students, considering academic, lifestyle, and institutional factors, from the students' own perspectives and environments."

3.3 Subgoals

1. Assess the impact of academic pressure on students' mental health.

2. Evaluate the role of personal responsibilities and lifestyle factors.
3. Explore students’ perception of mental health awareness and support systems.
4. Identify the top self-reported factors negatively affecting mental health.

3.4 Questions and Metrics

Subgoal 1: Assess the impact of academic pressure on students' mental health

Q1. How often do students feel overwhelmed by their academic workload?

- **M1.** Frequency scale response (Never, Sometimes, Often, Always) – *Qualitative*
- **M2.** Textual elaboration from free-response fields – *Qualitative*

Q2. How many hours do students spend on academic work daily?

- **M1.** Multiple-choice (e.g., "<3 hours", "3–5 hours", ">5 hours") – *Quantitative*

Q3. How clearly are academic deadlines and expectations communicated?

- **M1.** Subjective response on clarity scale – *Qualitative*

Subgoal 2: Evaluate the role of personal responsibilities and lifestyle factors

Q1. Do students have part-time or full-time jobs alongside their studies?

- **M1.** Binary response (Yes/No) – *Quantitative*

Q2. How far is the student's residence from their institution?

- **M1.** Categorical data (Less than 1 km, 1–5 km, etc.) – *Quantitative*

Q3. How much sleep do students typically get during academic terms?

- **M1.** Self-reported hours per night – *Quantitative*
- **M2.** Correlation with stress level – *Derived/Quantitative*

Q4. How does commuting or financial burden affect students’ stress levels?

- **M1.** Descriptive responses – *Qualitative*
- **M2.** Occurrence of financial-related issues in open-ended questions – *Qualitative*

Subgoal 3: Explore students' perception of mental health awareness and support

Q1. Do students think mental health should be part of the curriculum?

- **M1.** Yes/No – *Quantitative*
- **M2.** Justification provided – *Qualitative*

Q2. Are students willing to attend mental health workshops?

- **M1.** Response frequency (Always, Sometimes, Never) – *Qualitative*

Q3. Do students believe peer-led initiatives could reduce stigma?

- **M1.** Categorical opinion (Yes, No, Not Sure) – *Qualitative*

Q4. Have students used mental health apps or consulted professionals?

- **M1.** Binary responses (Yes/No for each) – *Quantitative*
- **M2.** Follow-up suggestions in open responses – *Qualitative*

Subgoal 4: Identify the top self-reported factors negatively affecting mental health

Q1. What societal factors (e.g., family expectations, social pressure) are commonly reported?

- **M1.** Keyword frequency in open responses – *Qualitative*

Q2. What personal challenges (e.g., procrastination, time management) appear repeatedly?

- **M1.** Recurring phrases in text analysis – *Qualitative*

Q3. What role does isolation or lack of peer connection play?

- **M1.** Mentions of loneliness or lack of support – *Qualitative*

Q4. What institutional improvements do students recommend?

- **M1.** Categorized suggestions (e.g., Counseling, Deadline policies) – *Qualitative*

- **M2.** Sentiment tone of open-ended suggestions – *Qualitative*

4. Questionnaire Preparation and Data Collection

4.1 Data Collection Methods

The data was collected through a Google Forms survey distributed among students of IIT DU. The form included both multiple-choice and open-ended questions targeting academic, personal, and societal factors influencing mental health. Participants provided responses anonymously to ensure honesty and confidentiality. A total of **35** responses were gathered during the month of January 2025. Proof of data collection includes timestamped form submissions and access to the raw spreadsheet file.

4.2 Tools Used

The primary tools used were:

- **Google Forms** – for creating and distributing the questionnaire.
- **Google Sheets / Excel** – for collecting and organizing the response data.
- **Manual review** – for extracting insights from open-ended responses.

*No automated tools or scripts were used in this version of the report for data visualization or analysis.

Some sample questions are given below-

1. What is your age?

- Below 18
- 18–20
- 21–23
- 24 and above

2. What is your gender?

- Male
- Female
- Non-binary
- Prefer not to say

3. Which year of study are you in?

- 1st Year
- 2nd Year
- 3rd Year
- 4th Year
- Master's

4. Where do you currently reside?

- On-campus hostel
- Off-campus accommodation
- At home

5. How far is your residence from your institution?

- Less than 1 km
- 1–5 km
- 6–10 km
- More than 10 km

6. How many hours do you spend on academic work daily?

- Less than 3 hours
- 3–5 hours
- More than 5 hours

7. How often do you feel overwhelmed by your academic workload?

- Never
- Sometimes
- Often
- Always

8. If given the option, would you attend mental health workshops regularly?

- Always
- Sometimes
- Not at all

9. Do you believe peer-led mental health initiatives would help reduce mental health stigma?

- Yes
- No
- Not sure

10. Do you think mental health should be included as part of the curriculum?

- Yes
- No

Full questionnaire can be found here:
https://docs.google.com/forms/d/1G03u588hNIfrPolIsPbTB_Jf6hwI0nWPj-gC_x03W2A/edit#responses

5. Data Visualization

The summary of the collected data is presented here in graphs and charts:

Demographic:

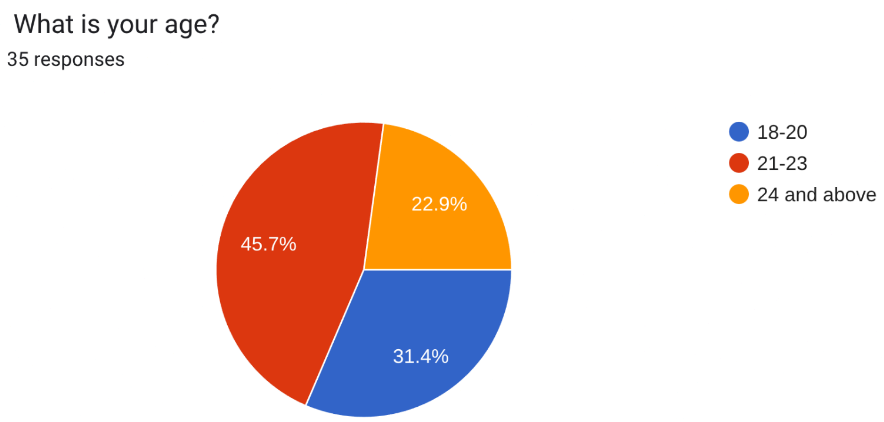

Fig 1: Respondents' age

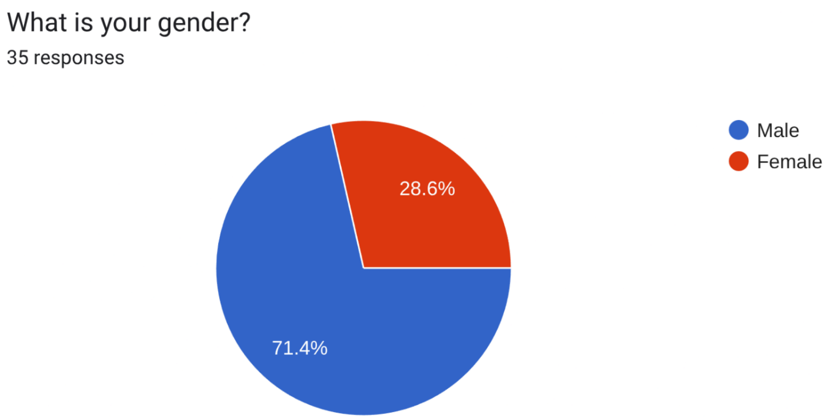

Fig 2: Respondents' gender

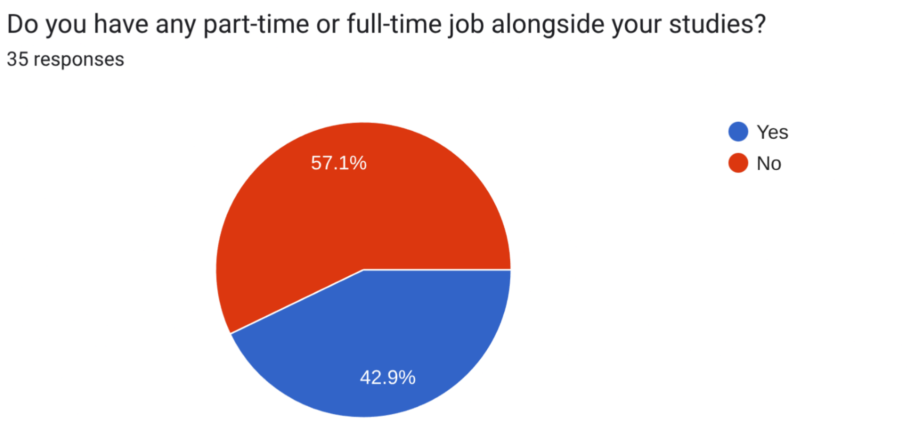

Fig 3: Respondents' residence

Academics:

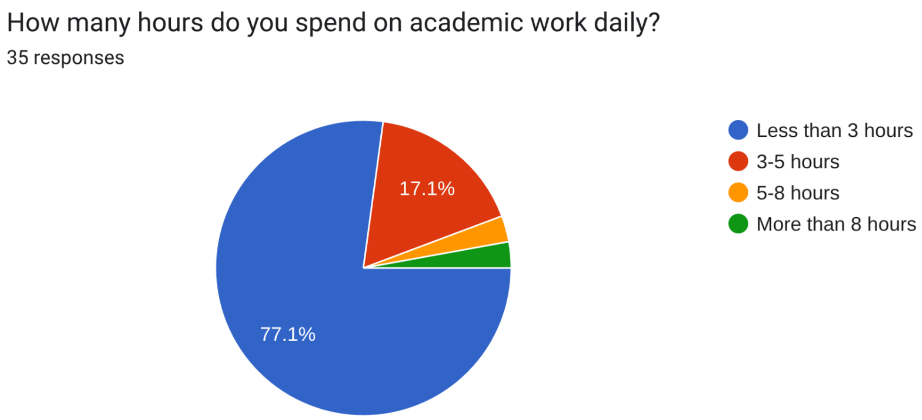

Fig 4: Respondents' daily study hours

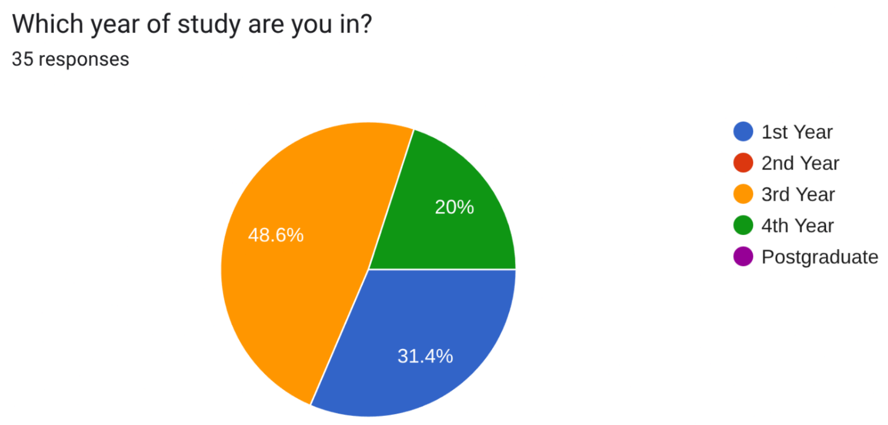

Fig 5: Respondents’ academic year

Mental load factors:

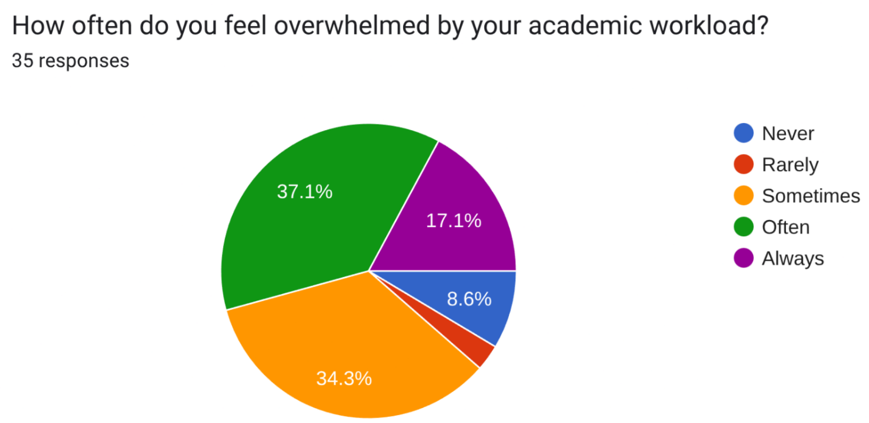

Fig 6: Respondents’ overwhelmness by academic workload

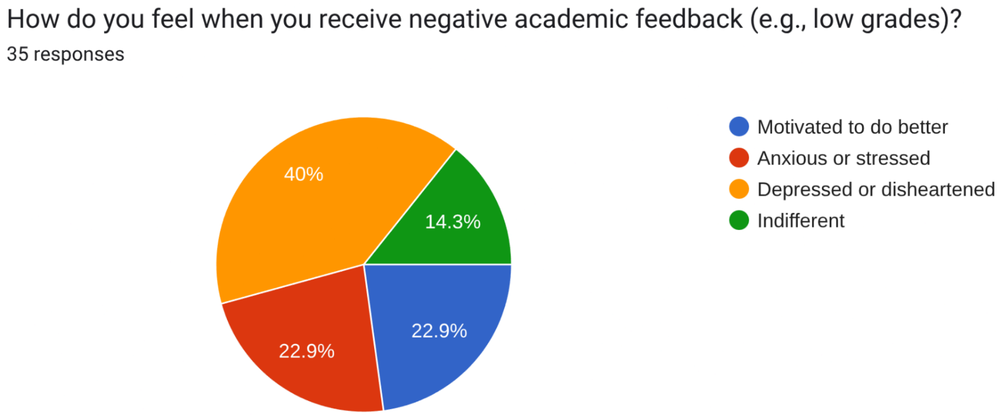

Fig 7: Respondents’ feeling after negative academic feedback

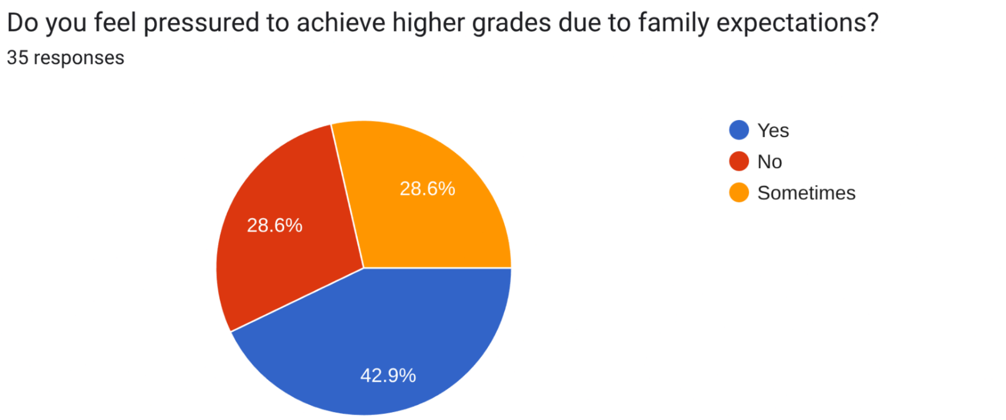

Fig 8: Respondents’ feeling of pressure due to family expectations

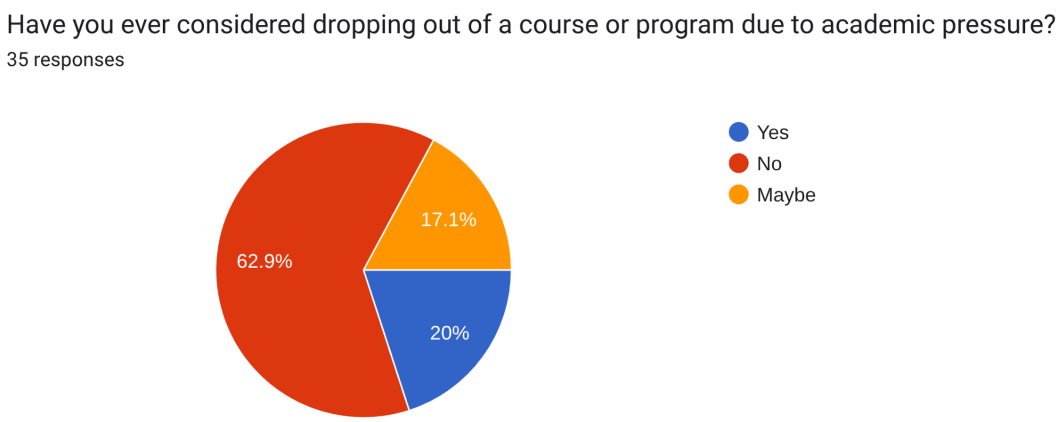

Fig 9: Respondents’ consideration of dropping out

Support Systems and Social Anxiety:

Do you have a support system (friends/family) that you can talk to about your problems?
35 responses

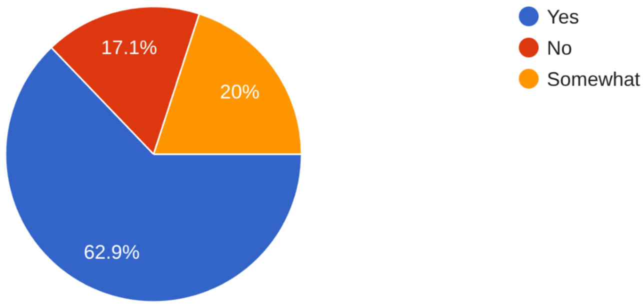

Fig 10: Respondents’ support system

Have you ever avoided social gatherings because of your mental health?
35 responses

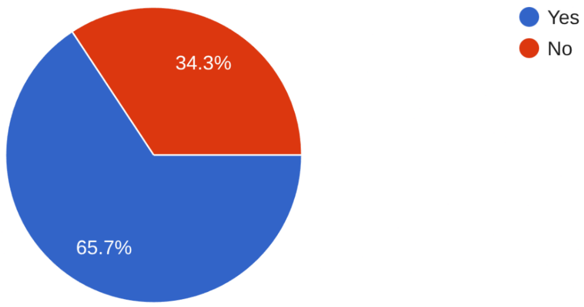

Fig 11: Respondents’ social anxiety

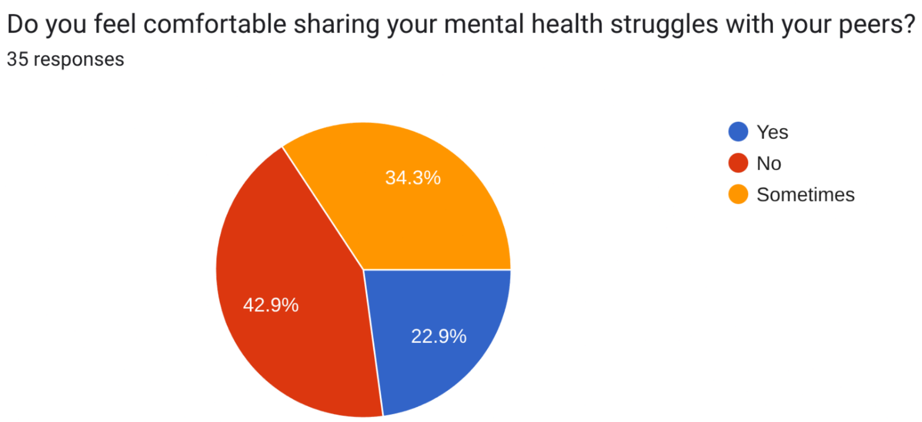

Fig 12: Respondents' ease of sharing mental health struggles

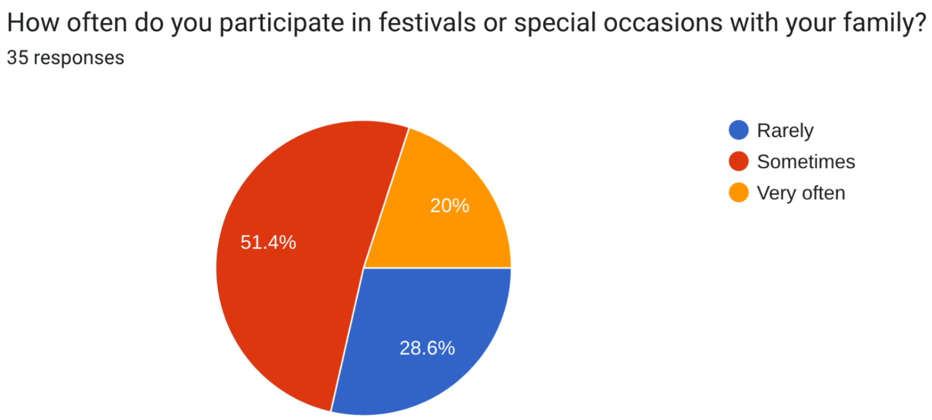

Fig 13: Respondents' participation in festivals and family occassions

Societal Impact:

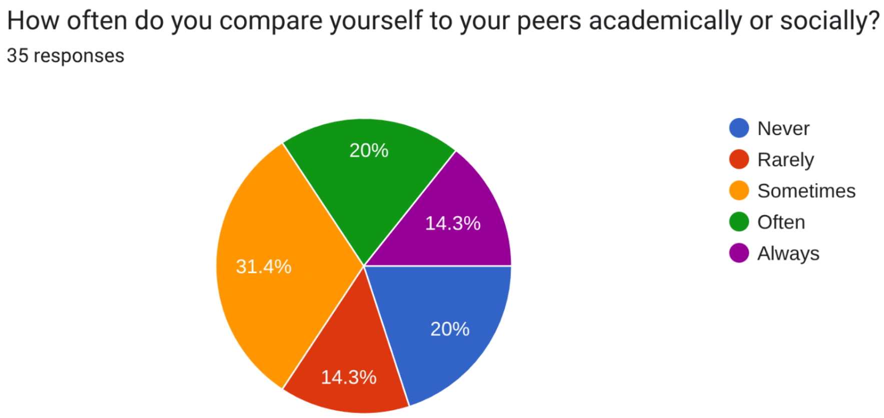

Fig 14: Respondents' academic and social comparisons

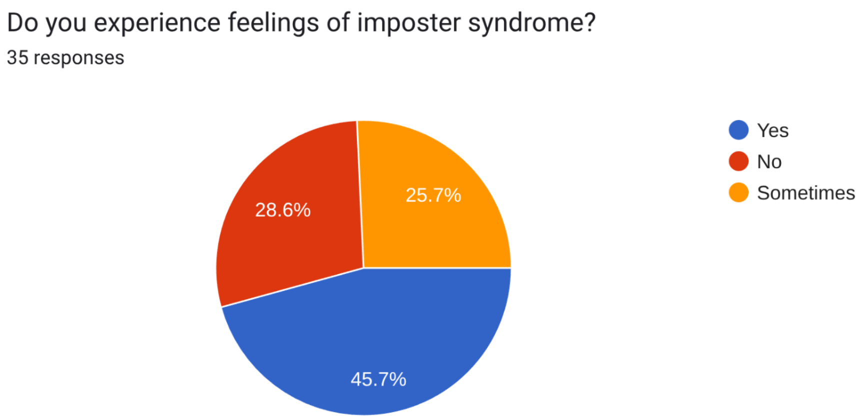

Fig 15: Respondents' feeling of imposter syndrome

Mental Health Situation:

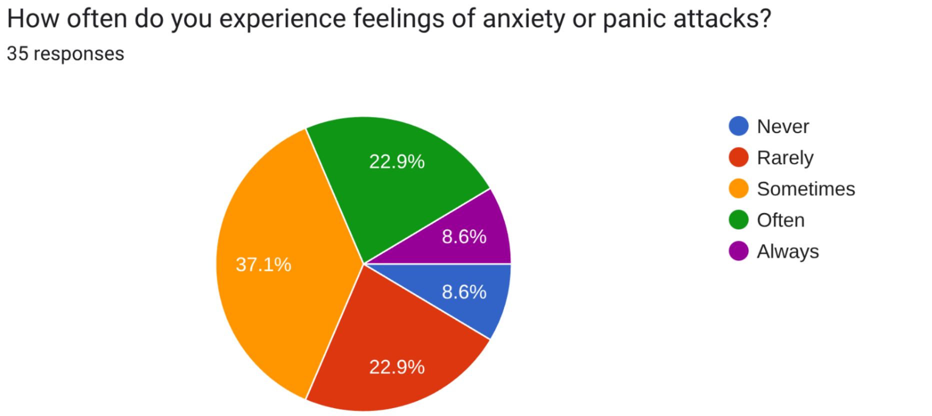

Fig 16: Respondents' feeling of anxiety

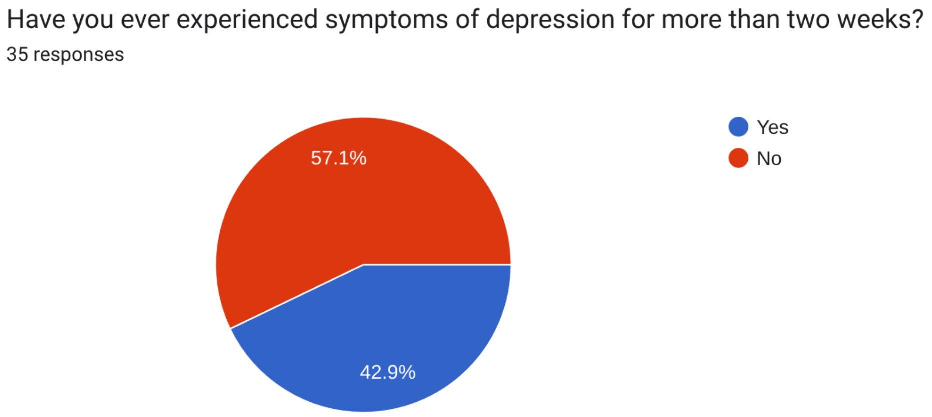

Fig 17: Respondents' symptoms of depression

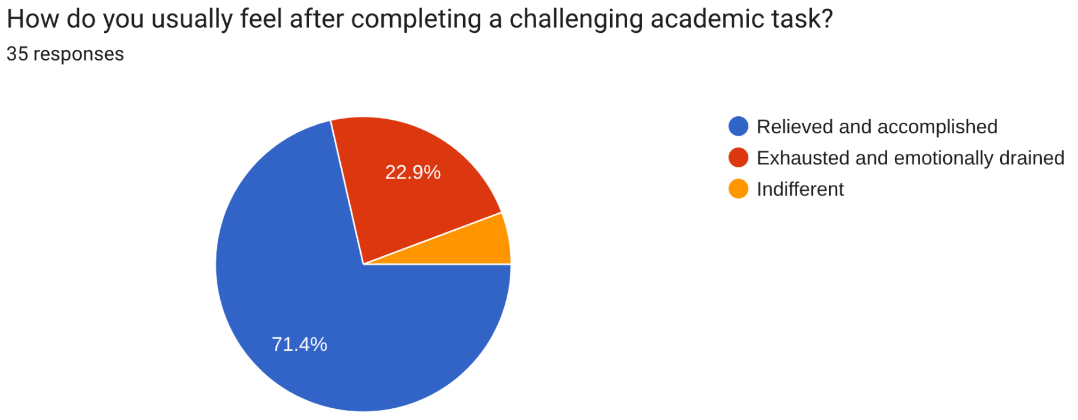

Fig 18: Respondents' feeling after completing challenging academic task

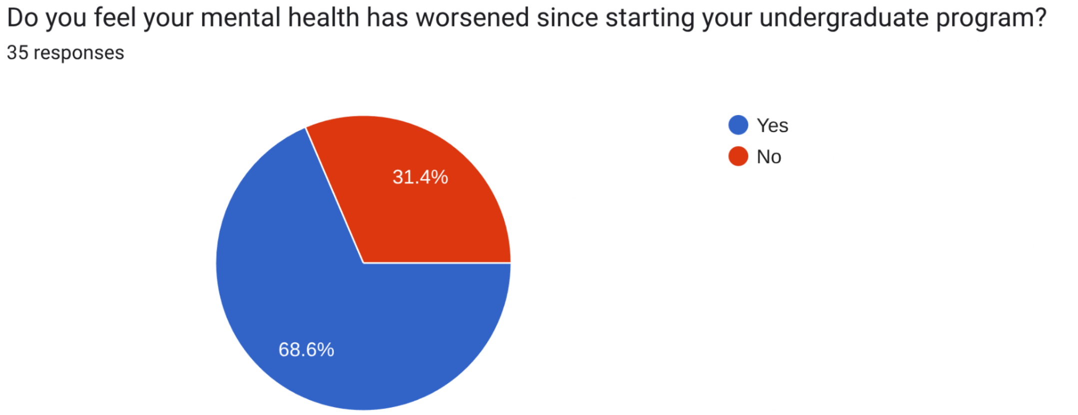

Fig 19: Respondents' current mental health situation

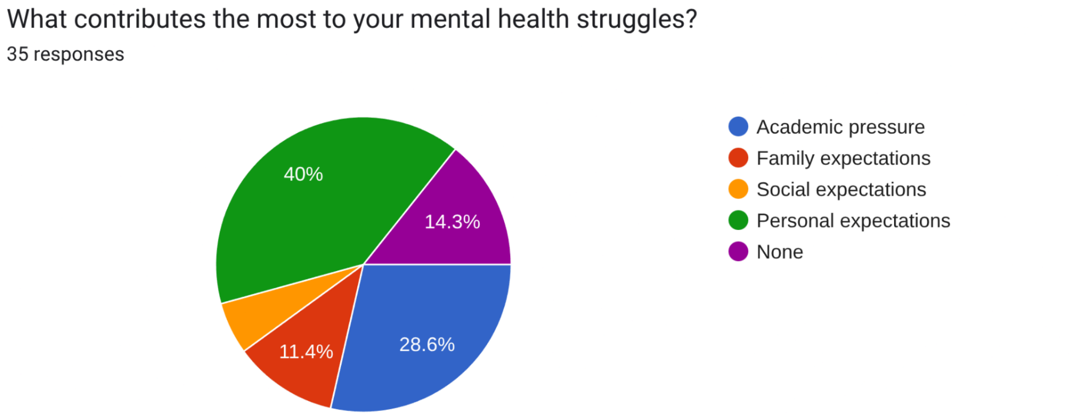

Fig 20: Contributors to respondents' mental halth struggles

Impact of Social Media and Technology:

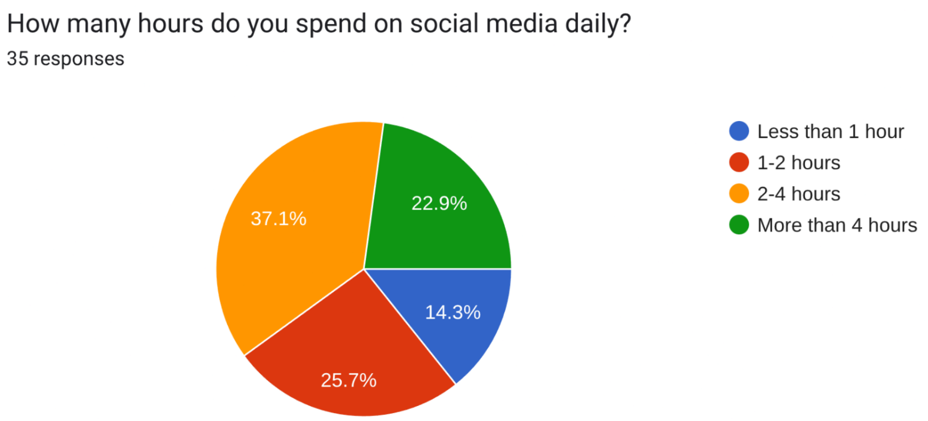

Fig 21: Respondents' daily social media usage

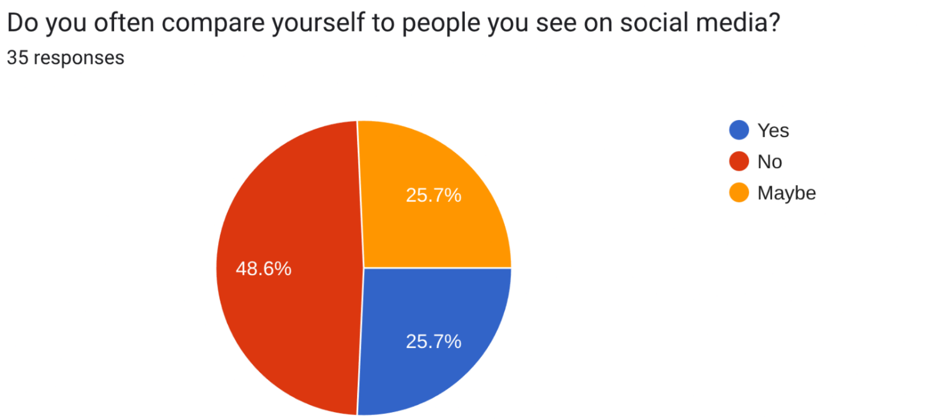

Fig 22: Respondents' self comparison to others on social media

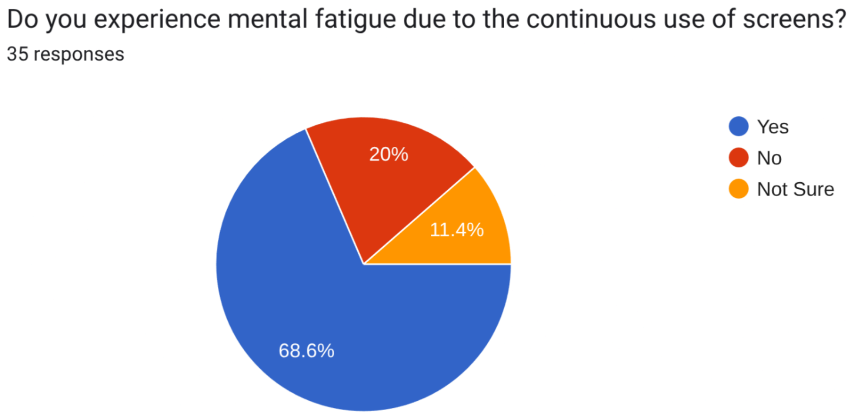

Fig 23: Impact of screen time on respondents' health

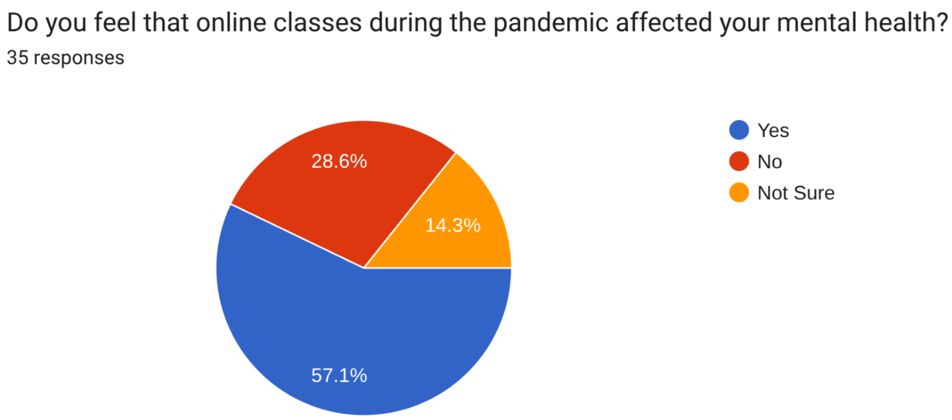

Fig 24: Respondents' opinion on online classes' impact on mental heallth

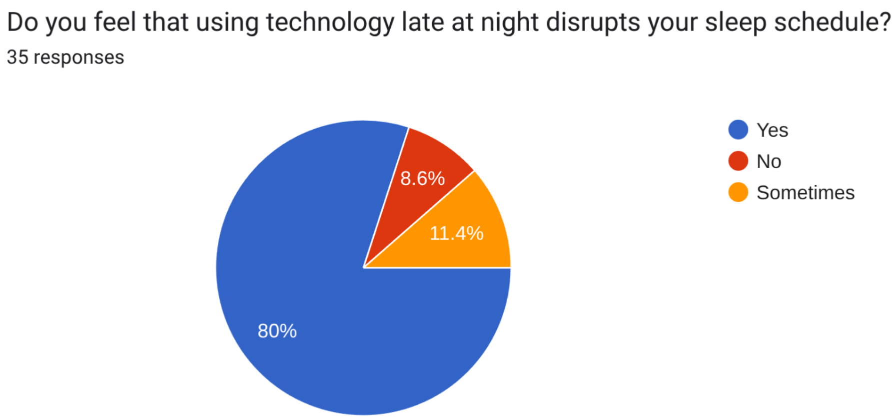

Fig 25: Impact of technology on Respondents’ sleep schedule

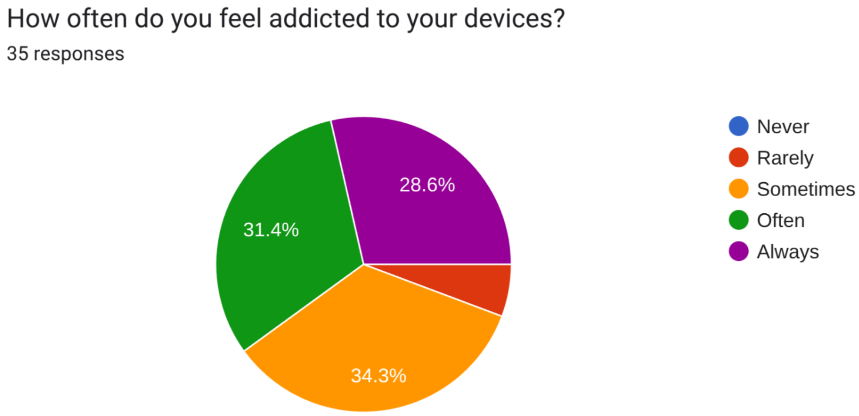

Fig 26: Respondents’ addiction to their devices

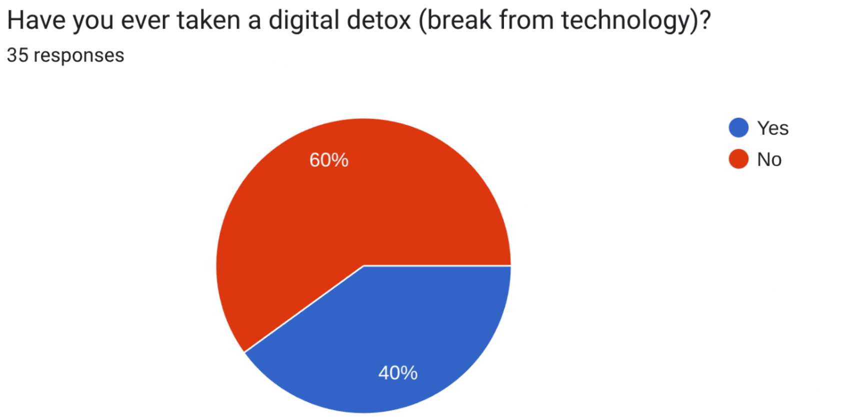

Fig 27: Respondents’ break from technology

Coping Mechanisms:

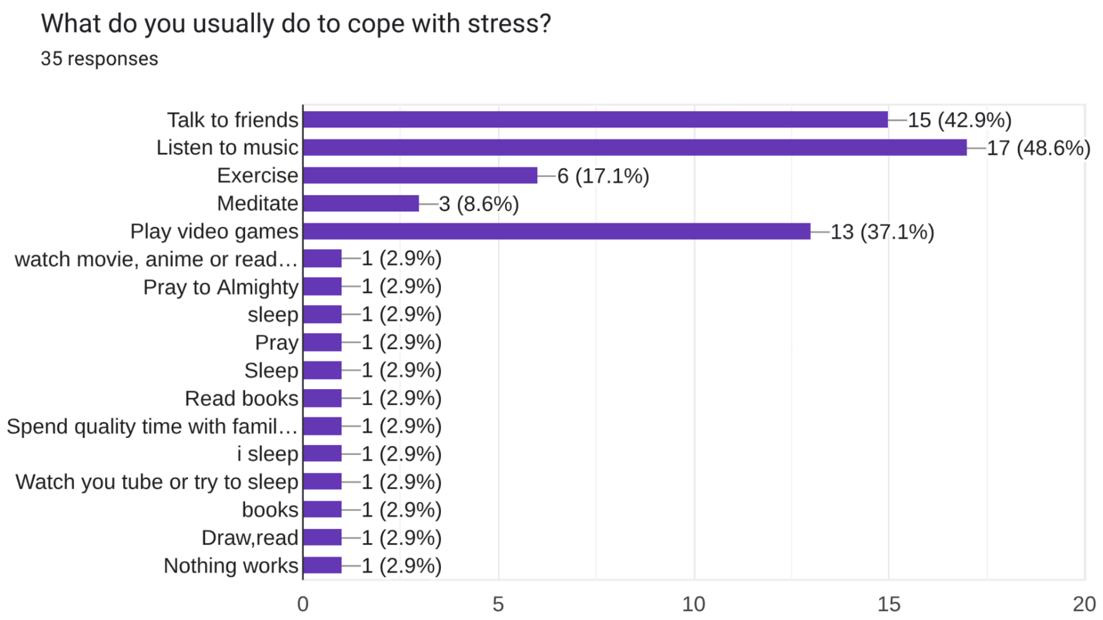

Fig 28: Respondents' coping mechanism for stress

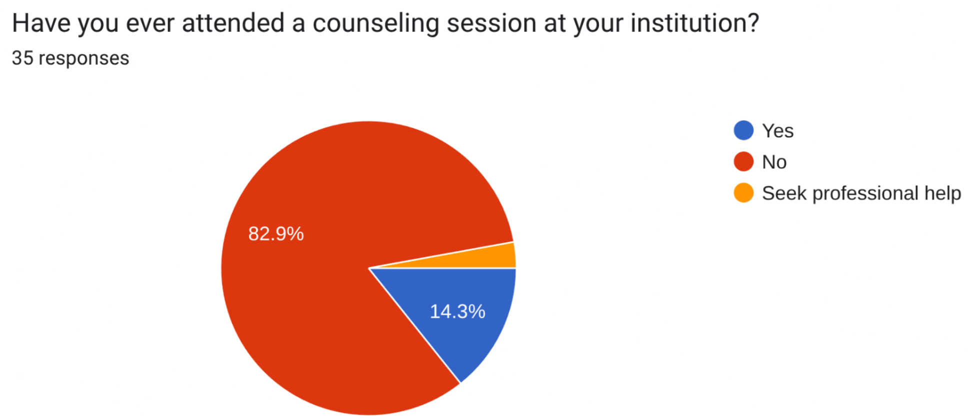

Fig 29: Respondents' attendance of counseling sessions

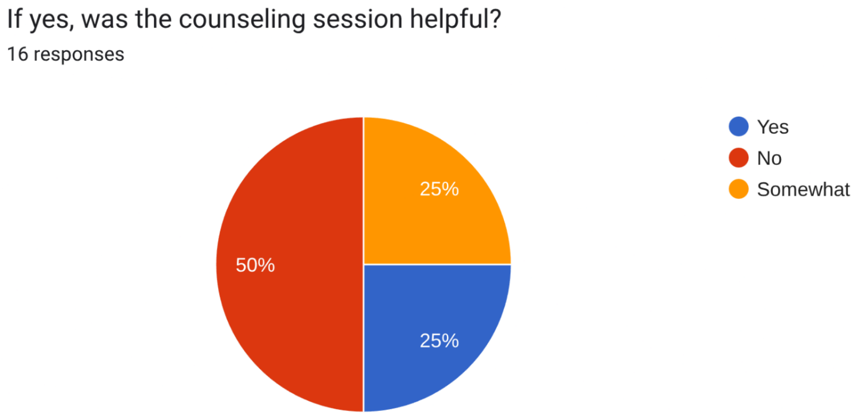

Fig 30: Respondents' opinion on counseling session

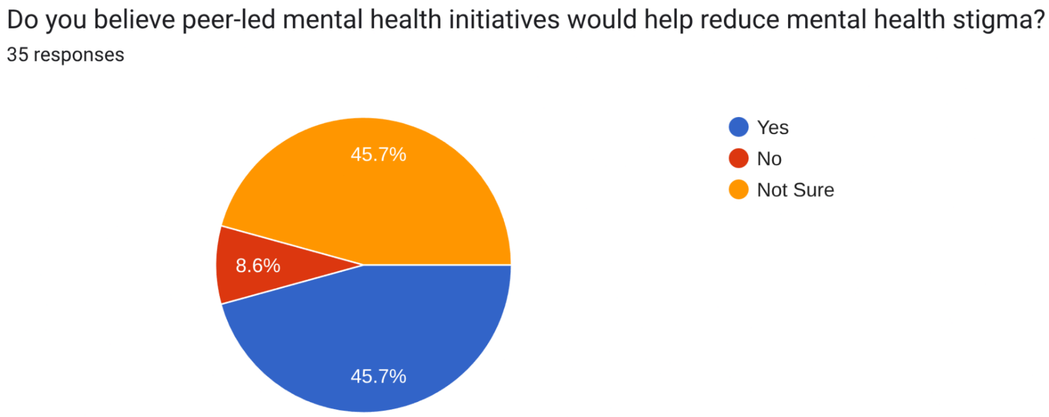

Fig 31: Respondents' opinion on peer-led health initiative

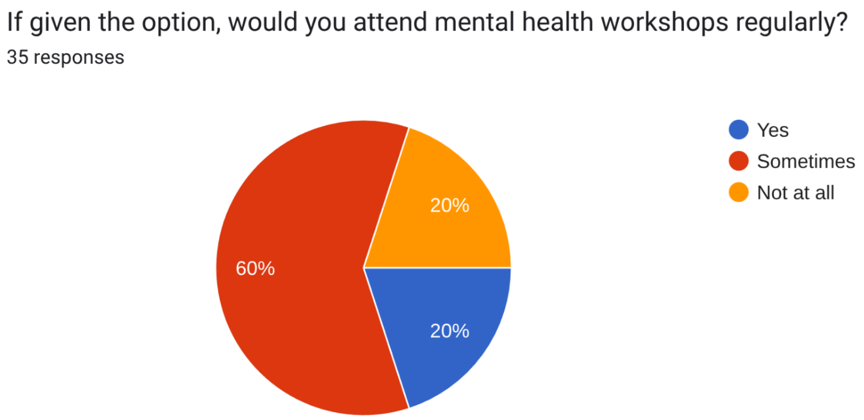

Fig 32: Respondents' opinion on attending mental health workshops

Current Mental Health Rating

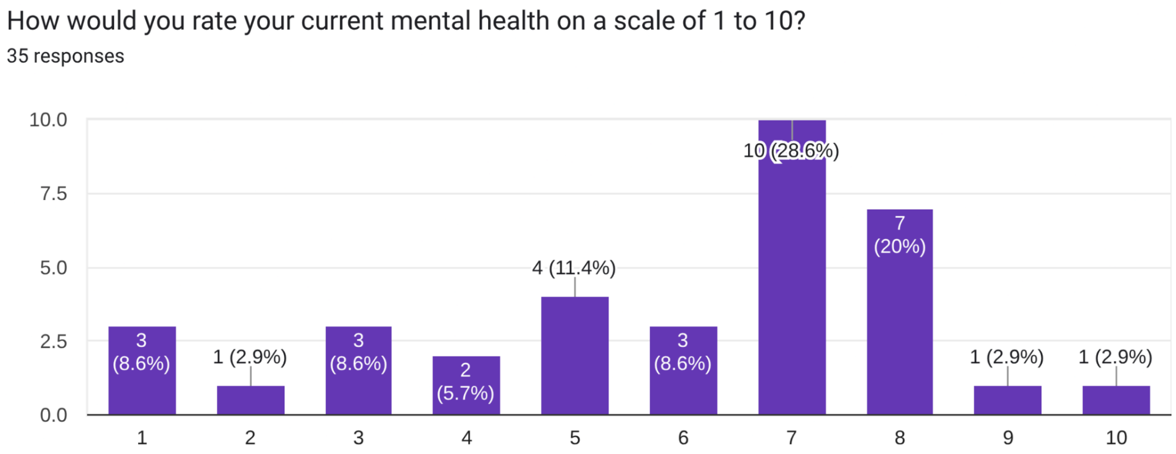

Fig 33: Respondents' rating on their current mental health

In few words, state how would you describe your overall mental health?

35 responses

Alhamdulillah for everything. Exam shesh hoile bachi.

f*cked up

Alhamdulillah

Overall I'm well Alhamdulillah, but sometimes my mental health get worsen due to Family problems.

Not so good

I'm doing good mentally but struggling a little bit regarding to my academic and career.

Worst in one word

At the moment, good

ok

Fig 34: Respondents' description of their overall mental health

6. Metrics Analysis

The survey conducted among IIT DU students provides a comprehensive dataset to evaluate societal and personal factors contributing to deteriorating mental health. The responses from 35 students highlight key patterns related to academic pressure, family expectations, social comparisons, and technology use, which are analyzed below through selected metrics.

Metric 1: Feeling Overwhelmed by Academic Workload

Hypothesis (H0): The proportion of students who frequently feel overwhelmed by their academic workload (responses "Often" or "Always") is equal to those who do not (responses "Never" or "Rarely"), ignoring "Sometimes" responses.
Observed Frequencies:
• Frequently Overwhelmed ("Often" or "Always"): 15 (Often) + 6 (Always) = 21
• Not Frequently Overwhelmed ("Never" or "Rarely"): 3 (Never) + 1 (Rarely) = 4
• Neutral ("Sometimes"): 10 (excluded from this analysis)
Expected Frequencies: If responses were evenly distributed between frequently and not frequently overwhelmed:
Expected Frequency = Frequently + Not Frequently
2 = 21 + 4
2 = 12.5
Chi-Square Test:
χ2 = ∑ (Observed − Expected)2
Expected
χ2 = (21 − 12.5)2
12.5 + (4 − 12.5)2
12.5 = 72.25
12.5 + 72.25
12.5 = 5.78 + 5.78 = 11.56

$$\chi^2 = \frac{(21-12.5)^2}{12.5} + \frac{(4-12.5)^2}{12.5} = \frac{72.25}{12.5} + \frac{72.25}{12.5} = 5.78 + 5.78 = 11.56$$

Degrees of freedom = 1. The p-value for χ2 = 11.56 is approximately 0.00067 (from chi-square distribution tables).

Interpretation: The p-value (0.00067) is less than the significance level (α = 0.05), leading to the rejection of the null hypothesis. This indicates that significantly more students feel frequently overwhelmed by their academic workload than those who do not. Academic pressure is a dominant factor contributing to mental health struggles, as 60% of respondents (21/35) reported feeling overwhelmed "Often" or "Always."

Metric 2: Pressure from Family Expectations

Hypothesis (H0): The proportion of students feeling pressured by family expectations to achieve higher grades (response "Yes") is equal to those who do not (responses "No" or "Sometimes").

Observed Frequencies:

• Pressured ("Yes"): 16

• Not Pressured ("No" or "Sometimes"): 8 (No) + 11 (Sometimes) = 19

Expected Frequencies:

Expected Frequency = 16 + 19

2 = 17.5

Chi-Square Test:

χ2 = (16 − 17.5)2

17.5 + (19 − 17.5)2

17.5 = 2.25

17.5 + 2.25

17.5 = 0.1286 + 0.1286 = 0.2572

$$\chi^2 = \frac{(16-17.5)^2}{17.5} + \frac{(19-17.5)^2}{17.5} = \frac{2.25}{17.5} + \frac{2.25}{17.5} = 0.1286 + 0.1286 = 0.2572$$

Degrees of freedom = 1. The p-value for χ2 = 0.2572 is approximately 0.612 (from chi-square distribution tables).

Interpretation: The p-value (0.612) is greater than α = 0.05, so we fail to reject the null hypothesis. This suggests no significant difference between students feeling pressured by family expectations and those who do not. However, 45.7% (16/35) of students reported "Yes" to family pressure, indicating it is a notable but not dominant factor compared to academic pressure.

Metric 3: Comparison to Peers on Social Media

Hypothesis (H0): The proportion of students who compare themselves to people on social media (responses "Yes" or "Maybe") is equal to those who do not (response "No").

Observed Frequencies:

• Compare ("Yes" or "Maybe"): 8 (Yes) + 9 (Maybe) = 17

• Do Not Compare ("No"): 18

Expected Frequencies:

Expected Frequency = 17 + 18

2 = 17.5

Chi-Square Test:

χ2 = (17 − 17.5)2

17.5 + (18 − 17.5)2

17.5 = 0.25

17.5 + 0.25

17.5 = 0.0143 + 0.0143 = 0.0286

$$\chi^2 = \frac{(17-17.5)^2}{17.5} + \frac{(18-17.5)^2}{17.5} = \frac{0.25}{17.5} + \frac{0.25}{17.5} = 0.0143 + 0.0143 = 0.0286$$

Degrees of freedom = 1. The p-value for $\chi 2$ = 0.0286 is approximately 0.866 (from chi-square distribution tables).

Interpretation: The p-value (0.866) is much greater than $\alpha = 0.05$, so we fail to reject the null hypothesis. This indicates no significant difference between students who compare themselves to others on social media and those who do not. Approximately 48.6% (17/35) engage in social media comparisons, suggesting it is a common but not statistically dominant factor affecting mental health.

Discussion

The analysis reveals that academic pressure is the most significant contributor to deteriorating mental health among IIT DU students, with 60% reporting frequent feelings of being overwhelmed. This is consistent with qualitative responses citing "academic pressure" and "deadlines" as top factors worsening mental health. Family expectations, while notable (45.7% feel pressured), do not show a statistically significant impact, suggesting that personal and institutional factors may outweigh societal expectations in this context. Social media comparisons, although prevalent, also lack statistical significance, indicating that while they contribute to mental health struggles, they are not a primary driver.
An anomaly observed is the high prevalence of "Relieved and accomplished" feelings after completing challenging tasks (27/35), despite many reporting negative emotions like anxiety or depression during the process. This suggests a complex emotional cycle where students experience stress but find relief upon task completion, potentially masking deeper mental health issues. Additionally, first-year students (9/35) reported lower mental health scores (average 4.33) compared to third-year (average 6.29) and fourth-year students (average 6.20), possibly due to the transition to university life.

Metric Effectiveness

The chosen metrics effectively captured key aspects of mental health stressors:

- Academic Workload: The metric on feeling overwhelmed was highly effective, as the significant p-value confirmed academic pressure as a primary factor. The clear distinction between "Often/Always" and "Never/Rarely" responses allowed for robust statistical analysis.
- Family Expectations: This metric was moderately effective. While it highlighted a notable proportion of students feeling pressured, the lack of statistical significance

suggests that combining "No" and "Sometimes" may have diluted the effect. Future surveys could use a Likert scale for finer granularity.

- Social Media Comparison: This metric was less effective due to the even distribution of responses, indicating that social media may not be a primary driver or that the question format did not capture nuanced behaviors. A more specific question (e.g., frequency of comparison) could improve effectiveness.

Overall, the metrics provided valuable insights but could be enhanced by incorporating additional questions on specific societal factors (e.g., peer pressure intensity) and using Likert scales for more precise data. The survey's qualitative responses, such as suggestions for counseling and reduced academic pressure, further validate the quantitative findings and suggest actionable institutional improvements.

7. Final Thoughts and Implications

The survey results highlight the increasing mental health challenges faced by students of IIT DU, driven by a combination of academic, personal, and societal stressors. While many students demonstrate resilience and self-awareness, the data reveals a significant portion of respondents frequently feel overwhelmed by their academic workload and struggle with time management, family expectations, and lack of sleep.

Students who live far from campus or juggle part-time jobs appear to carry additional burdens, which contribute to emotional strain. Interestingly, despite these challenges, most students rate their mental health between moderate to good on a self-assessed scale—suggesting a level of personal coping, yet still pointing to hidden or unspoken struggles.

The responses also show that awareness around mental health support remains mixed. While many students expressed interest in attending workshops or engaging in peer-led initiatives, a considerable number were unsure or had never sought professional help. This highlights a persisting stigma or lack of trust in mental health infrastructure.

Open-text responses provided meaningful insights into what students feel would improve their well-being, including clearer academic instructions, less exam pressure, and access to regular counseling sessions. The call for institutional reform was consistent—students want proactive support, not reactive responses.

Most notably, the survey exposed a shared understanding among students: **mental health matters and should be an integral part of the academic environment**. With many endorsing the idea of mental health checkups and curriculum inclusion, it's clear that institutional initiatives must evolve from optional resources to essential systems.

Overall, the findings underscore an urgent need for IIT DU to establish structured, empathetic, and inclusive support mechanisms. By doing so, the institution can not only improve student well-being but also create a healthier, more productive academic culture.